

Problem

Determine the Fourier transforms of these functions:

(a) $f(t) = 8t^2$

(b) $g(t) = 4(4 + t^2)$

Step-by-step solution

Step 1 of 4

(a)

The signal is,

$$f(t) = \frac{8}{t^2}$$

The Fourier transform of the signal $f(t)$ is obtained by applying the duality property of Fourier transform.

$$F(t) \xleftrightarrow{F} 2\pi f(-\omega)$$

Consider the Fourier transform pair,

$$|t| \xleftrightarrow{F} \frac{-2}{\omega^2}$$

By duality property,

$$\frac{-2}{t^2} \xleftrightarrow{F} 2\pi |-\omega|$$

$$\frac{2}{t^2} \xleftrightarrow{F} -2\pi |\omega| \dots\dots (1)$$

Step 2 of 4

Multiply equation (1) by 4 on both sides.

$$4\left(\frac{2}{t^2}\right) \xleftrightarrow{F} 4(-2\pi |\omega|)$$

$$\frac{8}{t^2} \xleftrightarrow{F} -8\pi |\omega|$$

Thus, the Fourier transform of the signal $f(t)$ is, $F(\omega) = -8\pi |\omega|$.

Step 3 of 4

(b)

The signal is,

$$g(t) = \frac{4}{4+t^2}$$

The Fourier transform of the signal $g(t)$ is obtained by applying the duality property of Fourier transform.

$$F(t) \xleftrightarrow{F} 2\pi f(-\omega)$$

Consider the Fourier transform pair,

$$e^{-a|t|} \rightarrow \frac{2a}{a^2 + \omega^2}$$

Step 4 of 4

Applying duality property,

$$\frac{2a}{a^2 + t^2} \xleftrightarrow{F} 2\pi e^{-a|\omega|}$$

$$\frac{2a}{a^2 + t^2} \xleftrightarrow{F} 2\pi e^{-a|\omega|}$$

Assume that $a = 4$.

Hence,

$$\frac{4}{4+t^2} \rightarrow 2\pi e^{-2|\omega|}$$

Thus,

$$F\left(\frac{4}{4+t^2}\right) = 2\pi e^{-2|\omega|}$$

Thus, the Fourier transform of the signal $g(t)$ is $G(\omega) = 2\pi e^{-2|\omega|}$.